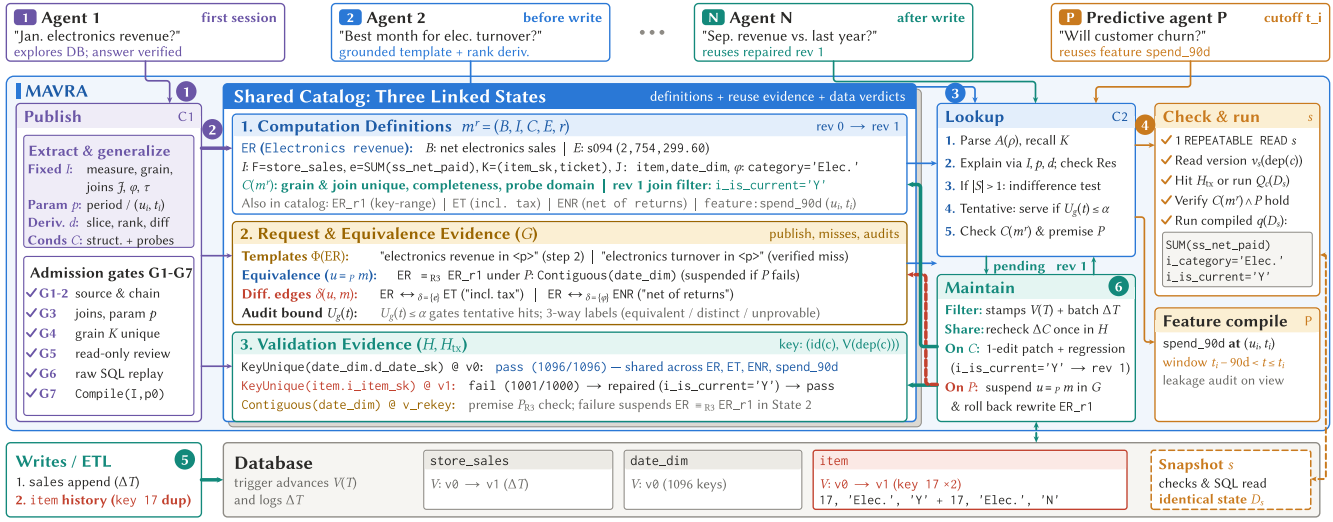

1 Agent 1
first session
"Jan. electronics revenue?"
explores DB; answer verified
2 Agent 2
before write
"Best month for elec. turnover?"
grounded template + rank deriv.
N Agent N
after write
"Sep. revenue vs. last year?"
reuses repaired rev 1
P Predictive agent P
cutoff t_i
"Will customer churn?"
reuses feature spend_90d
MAVRA
1
Publish C1
Extract & generalize
Fixed I: measure, grain, joins J, φ, τ
Param p: period / (u_i, t_i)
Deriv. d: slice, rank, diff
Conds C: struct. + probes
Admission gates G1-G7
✓G1-2 source & chain
✓G3 joins, param p
✓G4 grain K unique
✓G5 read-only review
✓G6 raw SQL replay
✓G7 Compile(I,p0)
2
Shared Catalog: Three Linked States
definitions + reuse evidence + data verdicts
1. Computation Definitions m^r = (B, I, C, E, r)
rev 0 → rev 1
ER (Electronics revenue): B: net electronics sales | E: s094 (2,754,299.60)
I: F=store_sales, e=SUM(ss_net_paid), K=(item_sk, ticket), J: item, date_dim, φ: category='Elec.'
C(m^r): grain & join unique, completeness, probe domain | rev 1 join filter: i_is_current='Y'
Also in catalog: ER_r1 (key-range) | ET (incl. tax) | ENR (net of returns) | feature:spend_90d (u_i, t_i)
2. Request & Equivalence Evidence (G)
publish, misses, audits
Templates Φ(ER): "electronics revenue in <p>" (step 2) | "electronics turnover in <p>" (verified miss)
Equivalence (u =_P m): ER =_R3 ER_r1 under P: Contiguous(date_dim) (suspended if P fails)
Diff. edges δ(u, m): ER ↔ δ={φ} ET ("incl. tax") | ER ↔ δ={φ} ENR ("net of returns")
Audit bound U_g(t): U_g(t) ≤ α gates tentative hits; 3-way labels (equivalent / distinct / unprovable)
3. Validation Evidence (H, H_tx)
key: (id(c), V(dep(c)))
KeyUnique(date_dim.d_date_sk) @ v0: pass (1096/1096) — shared across ER, ET, ENR, spend_90d
KeyUnique(item.i_item_sk) @ v1: fail (1001/1000) → repaired (i_is_current='Y') → pass
Contiguous(date_dim) @ v_rekey: premise P_R3 check; failure suspends ER =_R3 ER_r1 in State 2
3
Lookup C2
1. Parse A(ρ), recall K
2. Explain via I, p, d; check Res
3. If |S| > 1: indifference test
4. Tentative: serve if U_g(t) ≤ α
5. Check C(m^r) & premise P
pending rev 1
Maintain 6
Filter: stamps V(T) + batch ΔT
Share: recheck ΔC once in H
On C: 1-edit patch + regression (i_is_current='Y' → rev 1)
On P: suspend u =_P m in G & roll back rewrite ER_r1
4
Check & run s
✓ 1 REPEATABLE READ s
✓ Read version v_s(dep(c))
✓ Hit H_tx or run Q_s(D_s)
✓ Verify C(m^r) ∧ P hold
✓ Run compiled q(D_s):
SUM(ss_net_paid)
i_category='Elec.'
i_is_current='Y'
Feature compile P
spend_90d at (u_i, t_i)
window t_i − 90d < t ≤ t_i
leakage audit on view
Writes / ETL 5
1. sales append (ΔT)
2. item history (key 17 dup)
Database
trigger advances V(T) and logs ΔT
store_sales
V: v0 → v1 (ΔT)
date_dim
V: v0 (1096 keys)
item
V: v0 → v1 (key 17 ×2)
17, 'Elec.', 'Y' + 17, 'Elec.', 'N'
Snapshot s
checks & SQL read
identical state D_s